

Project 2

By

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Introduction to Analytics

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Part 1

Title: Exploratory Data Analysis on World Happiness and Freedom (2015)

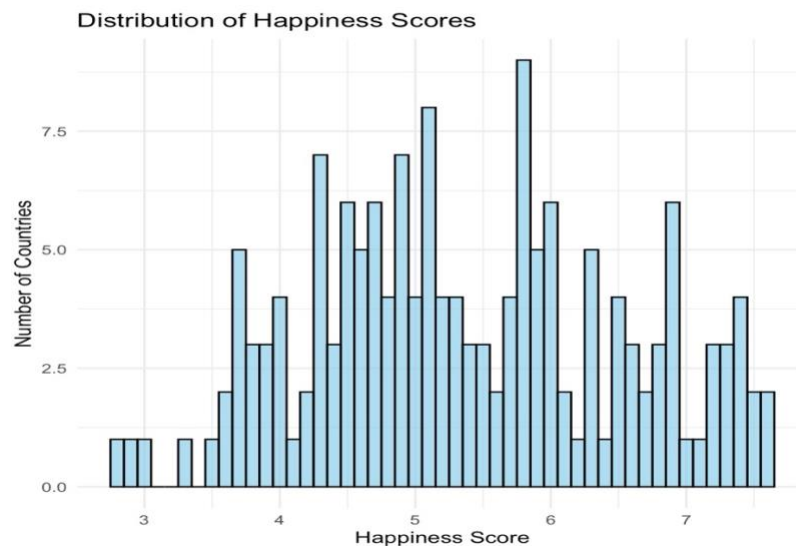
1. Introduction

In this report, I explored the 2015 World Happiness dataset to better understand how happiness levels vary around the world and how they relate to people's sense of freedom. I investigated which countries had the highest and lowest freedom scores and compared average happiness across different global regions. To make things clear and easy to follow, I included the R code I used, along with the results and visual summaries.

The dataset includes key variables such as happiness score, freedom score, economy (GDP per capita), family/social support, and health (life expectancy) for each country.

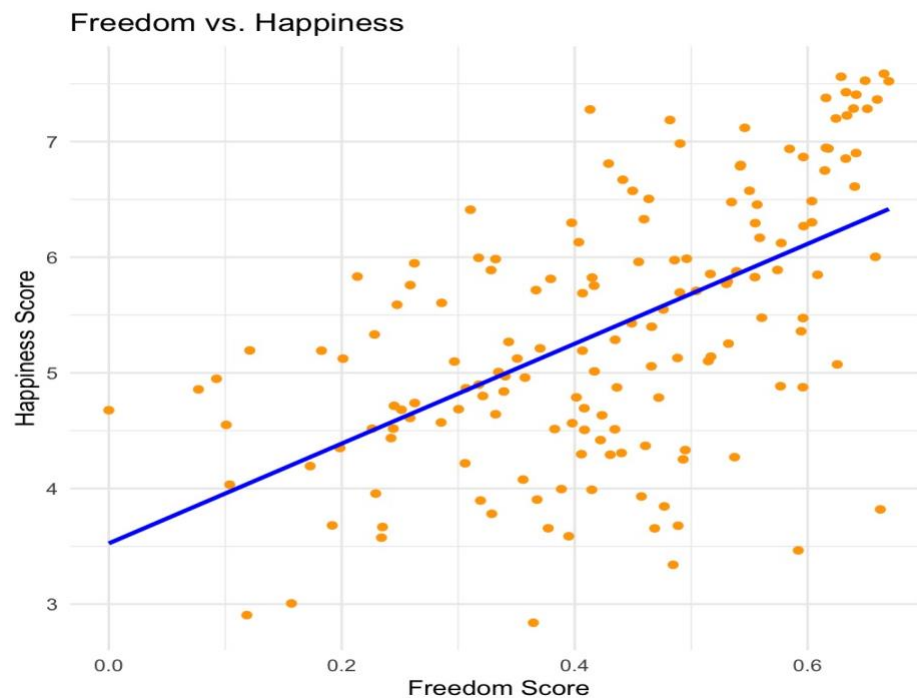
2. Key Findings

- **Happiness Score Distribution**



A histogram was created to visualize the distribution of happiness scores across countries. The chart revealed that most countries had happiness scores clustered between 4 and 7. While a few countries exhibited significantly higher happiness scores, the majority fell within a moderate range. This suggests that most countries experience a level of happiness that is neither exceptionally high nor low, with only a few countries reporting very high levels of well-being.

- **The Relationship Between Freedom and Happiness**



The analysis also examined the relationship between freedom scores and happiness. A scatter plot illustrated a positive correlation between these two variables: countries with higher freedom scores tended to have higher happiness scores. The linear regression line added to the plot supported this trend, indicating that greater political and personal freedoms are associated with higher overall happiness in these countries.

- **Top 10 Happiest Countries**

Table: Top 10 Countries from Happiness Dataset (2015)

country	region	happiness_score	freedom
Switzerland	Western Europe	7.587	0.66557
Iceland	Western Europe	7.561	0.62877
Denmark	Western Europe	7.527	0.64938
Norway	Western Europe	7.522	0.66973
Canada	North America	7.427	0.63297
Finland	Western Europe	7.406	0.64169
Netherlands	Western Europe	7.378	0.61576
Sweden	Western Europe	7.364	0.65980
New Zealand	Australia and New Zealand	7.286	0.63938
Australia	Australia and New Zealand	7.284	0.65124

To highlight the leading countries in terms of happiness, the top 10 happiest countries were identified based on their happiness scores. Countries like Switzerland, Iceland, and Denmark consistently ranked high for both happiness and freedom. These countries generally have strong social support systems, good governance, and economic stability, factors that significantly contribute to the well-being of their citizens. The analysis confirmed that these factors play a crucial role in fostering high levels of happiness.

- **Low Freedom and Happiness Countries**

On the other hand, the analysis also identified countries with the lowest freedom scores, which were associated with lower happiness scores. For example, countries such as Syria and Afghanistan, which ranked poorly in both freedom and happiness, highlighted the importance of personal freedoms and civil rights in determining happiness. The data suggests that when individuals lack freedom, their overall happiness tends to be significantly lower.

3. Key Visualizations Supporting the Findings

Several visualizations were employed to enhance the understanding of the findings:

- **Histogram of Happiness Scores:** This graph provided a clear representation of the distribution of happiness across countries.
- **Scatter Plot of Freedom vs. Happiness:** This visualization demonstrated the positive correlation between freedom and happiness.
- **Top 10 Countries by Happiness:** The table listing the top 10 happiest countries highlighted the global leaders in terms of happiness.

4. Recommendations

Based on the findings from this analysis, several recommendations can be made to improve global happiness levels:

- **Enhance Political and Personal Freedoms:**

The analysis suggests a strong connection between freedom and happiness. Countries can foster greater happiness by improving political transparency, safeguarding civil liberties,

and ensuring citizens' freedom to make personal choices. Encouraging individual autonomy can significantly boost the overall well-being of the population.

- **Strengthen Social Support Systems:**

Countries that have strong social support systems such as healthcare, education, and social safety nets—tend to have higher happiness scores. Governments should prioritize investing in these systems to ensure that citizens feel supported and secure. This is particularly important in times of crisis or personal hardship, where access to support can make a significant difference in well-being.

- **Promote Equitable Economic Growth:**

Economic prosperity alone is not enough to ensure happiness. It is important that growth is inclusive and that the benefits of economic development reach all sections of society. Governments should focus on reducing income inequality and promoting sustainable, inclusive economic policies that contribute to long-term well-being.

- **Invest in Education and Mental Health:**

Education and mental health support are crucial to increasing happiness. Providing access to quality education and offering mental health programs can help individuals lead fulfilling lives. Mental health initiatives, in particular, should be integrated into national healthcare systems to address the psychological well-being of citizens.

- **Foster Strong Communities:**

A sense of belonging and community is key to individual happiness. Governments and organizations should encourage social cohesion by creating spaces where people can connect, engage, and collaborate. Building strong communities can help individuals feel supported and less isolated, thus contributing to higher happiness levels.

5. Conclusion

After analyzing the 2015 World Happiness dataset, several key insights into global happiness trends and its correlation with freedom have emerged. The data highlights a clear relationship between personal freedoms and happiness: countries with higher levels of freedom tend to have higher happiness scores. This suggests that freedoms such as civil liberties, political rights, and individual autonomy significantly contribute to a population's overall well-being.

The analysis also identified regional patterns, with Scandinavian countries like Denmark, Finland, and Norway, leading in happiness rankings. These nations have robust social support systems and high freedom scores, indicating that such factors are instrumental in enhancing happiness. Conversely, countries with low freedom scores, such as Syria and Afghanistan, have some of the lowest happiness scores, suggesting that restricted freedoms can negatively affect people's happiness and overall quality of life.

Part 2

Title : Exploratory Data Analysis of Major League Baseball (MLB) Batting Statistics (1986)

1. Introduction

The primary focus of this analysis was to explore the 1986 Major League Baseball (MLB) batting statistics. The objective was to understand the performance of players in key areas such as batting average (BA), on-base percentage (OBP), home runs (HR), and runs batted in (RBI). By analyzing this data, I aimed to identify key trends, relationships between different statistics, and evaluate the top performers of the season. Additionally, visualizations were employed to highlight patterns, making it easier to interpret the data and draw conclusions from it.

2. Key Findings

- **Top Performers:**

The analysis revealed a list of standout players based on their batting statistics. Notable players like Tony Gwynn showcased an impressive batting average, while power hitters such as Mark McGwire and Jose Canseco excelled in home runs. These players stood out for their contributions to their teams through consistent batting and power hitting.

- **Correlation Between Batting Average and Home Runs:**

A negative correlation was observed between batting average and home runs. Players who focused on hitting home runs (i.e., power hitters) often had lower batting averages, which suggests that hitting for power may come at the cost of batting consistency.

- **Top MVP Candidates:**

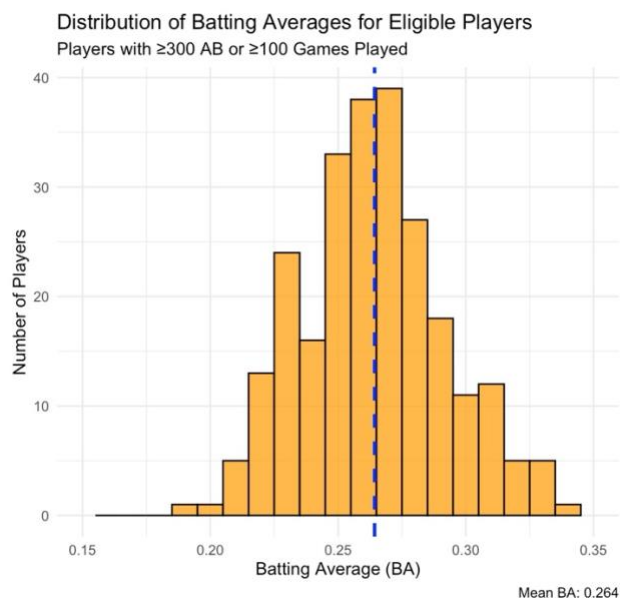
Through a ranking of key statistics OBP, HR, and RBI, the top candidates for the MVP award were identified. These players not only had strong performances in one area but also maintained solid contributions in other key categories, making them well-rounded assets to their teams.

- **Batting Average Distribution:**

The histogram of batting averages clearly showed how most players fell into the lower range of batting averages, between .250 and .280. Only a smaller group of players had batting averages above .300, demonstrating the difficulty of maintaining high averages in MLB.

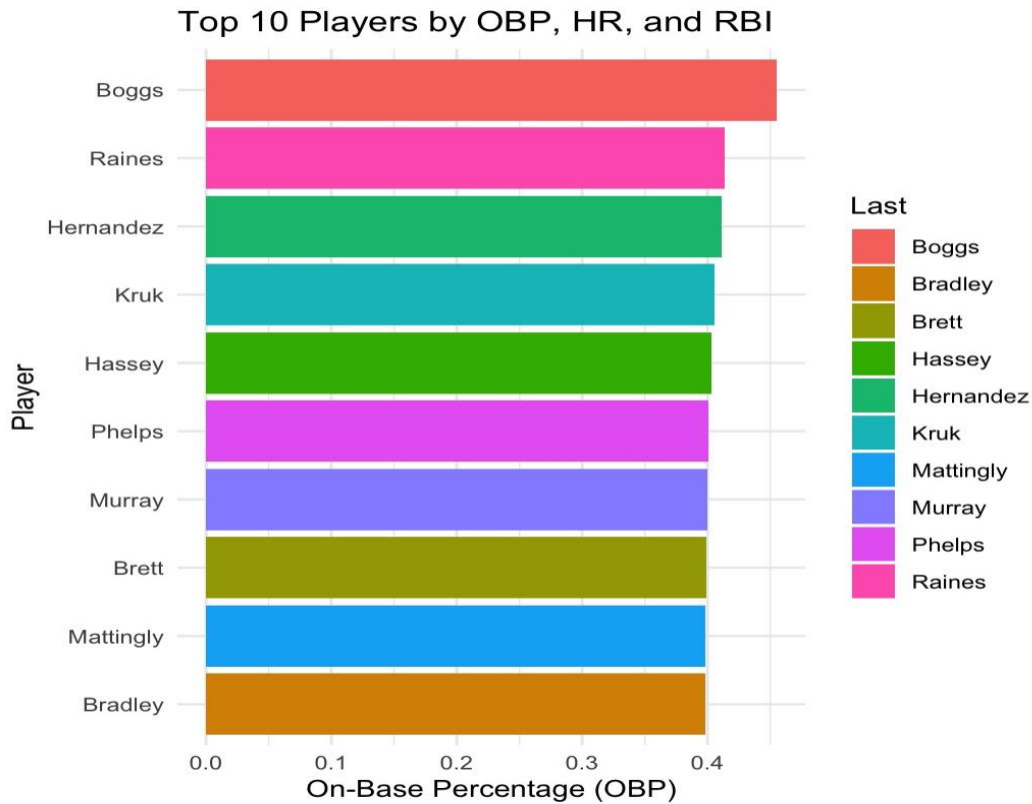
3. Data Visualizations

- **Distribution of Batting Averages:**



A histogram was used to illustrate the distribution of batting averages for players with at least 300 at-bats or 100 games played. This helped to identify the typical performance range for players and pinpoint outliers, players who performed exceptionally well or poorly. By visualizing the distribution of batting averages, it became clear where the majority of players fell on the spectrum. It also helped to spot extreme values (outliers) that might be worthy of further investigation, particularly players who had an exceptionally high or low performance in batting.

- **Top 10 Players by OBP, HR, and RBI:**



A bar chart was generated to display the top 10 players based on their on-base percentage (OBP), home runs (HR), and runs batted in (RBI). This allowed for a direct comparison of these players and highlighted those who were performing at the highest level across multiple key statistics. The bar chart made it easier to compare player performance in terms of OBP, HR, and RBI. By visualizing the top players, the chart helped to highlight the best all-around performers and provided a clear way to identify potential MVP candidates.

4. Recommendations:

- **Focus on Balanced Player Development:**

Teams should consider fostering balanced skill sets in players, focusing not just on power hitting but also on maintaining a solid batting average. Developing players who can consistently hit for average while still contributing in home runs would be ideal.

- **Improve Consistency in Batting:**

While power hitters are valuable, teams should also emphasize the importance of contact hitting and consistency. Batting coaches should work on improving players' ability to make consistent contact with the ball to help elevate overall team performance.

- **Utilize Data for Performance Monitoring:**

Teams should leverage advanced statistics to monitor and assess player performance more effectively. Metrics like OBP, exit velocity, and launch angle could provide more granular insights into player strengths and areas of improvement.

5. Conclusion:

The 1986 MLB batting statistics provided valuable insights into player performance, particularly in terms of key metrics like batting average, home runs, and on-base percentage. By analyzing these statistics and visualizing the data, several trends were uncovered, including the trade-off between batting average and home runs. It was evident that power hitters, while contributing significantly in home runs, often had lower batting averages. Additionally, the top players who excelled across multiple categories were identified, with several candidates standing out for their well-rounded performance.

References

Huynh, Y. W. (2019). Guide to R. *Bookdown*. https://bookdown.org/yih_huynh/Guide-to-R-Book/

Kubatko, J. (2016). StatMuse baseball statistics glossary. *StatMuse*. <https://blog.statmuse.com/statmuse-baseball-statistics-glossary-7be2a125d024/>

Urin1. (n.d.). Baseball statistics analysis using R. *RPubs*. <https://rpubs.com/urin1/852924>

Stat-at-ease. (2023). Baseball statistics explained | Learn R through Baseball (DataCamp tutorial) [Video]. YouTube. https://www.youtube.com/watch?v=dgPH_y6GwuY